

# Dr Moslem Uddin

Research Associate | UNSW Canberra | Academic CV

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## Research profile

Microgrid energy management, renewable integration, battery storage, resilience and intelligent power systems.

## Appointments

Research Associate, UNSW Canberra, September 2025-present.

Research Assistant, UNSW Canberra, March-August 2025.

Tutor and Lab Demonstrator, Universiti Teknologi PETRONAS, July 2016-February 2019.

## Education

PhD in Electrical Engineering (Power Systems), UNSW Canberra, 2025.

MSc by Research in Power Systems, Universiti Teknologi PETRONAS, 2019.

BSc in Electrical and Electronic Engineering, Prime University, 2014; Chancellor's Gold Medal.

## Selected publications

Microgrids: A review, outstanding issues and future trends. Energy Strategy Reviews, 2023.

Techno-economic potential of multi-energy community microgrid: The perspective of Australia. Renewable Energy, 2023.

A novel peak shaving algorithm for islanded microgrid using battery energy storage system. Energy, 2020.

A review on peak load shaving strategies. Renewable and Sustainable Energy Reviews, 2018.

Real-Time Energy Management Strategies for Community Microgrids. IEEE ETFG, 2025.

## Research and service

Community microgrid design; techno-economic analysis; supervisory and learning-based energy management.

Review Editor, Frontiers in Energy Research (2022-2025); Newsletter Editor, IEEE ACT Section (2024-present).

Tools: MATLAB/Simulink, Python, HOMER Pro, DigSILENT PowerFactory, PSS/E and PSCAD.

## Full publication record

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